

Maintenance Schedule

Diesel Engine

12V1600R50

Application Group 2A

MS50141/03E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
12V1600R50	58 kW/cyl.	2A, Continuous operation, unrestricted

Table 1: Applicability

Table of Contents

1	Maintenance Schedule		
1.1	Preface	4	
1.2	Correlation matrix for application group 2A	5	
1.3	Maintenance schedule matrix 30,000 h	6	
1.4	Maintenance tasks 30,000 h	8	
1.5	Maintenance schedule matrix 27,000 h	9	
1.6	Maintenance tasks 27,000 h		11
1.7	Maintenance schedule matrix 24,000 h		12
1.8	Maintenance tasks 24,000 h		13
1.9	Maintenance schedule matrix 21,000 h		14
1.10	Maintenance tasks 21,000 h		15

1 Maintenance Schedule

1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance specifications can result in a violation of the statutory emission regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be maintained, exchanged or repaired if the components used for this purpose are approved by MTU or equivalent components. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps and all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability. The maintenance intervals and required maintenance scopes are based on operational experience and must therefore be regarded as recommendations. In the event of difficult operational and ambient conditions, additional maintenance tasks and/or changes to the maintenance intervals may be necessary.

Maintenance activities are specified with intervals based on hours in operation and time limits. The limit which occurs first is applicable.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational checks and maintenance work that does not require disassembly of the product.

QL2: Component replacement (corrective, and thus not an element of the maintenance schedule).

QL3: Maintenance work that requires partial disassembly of the product.

QL4: Maintenance work that requires complete disassembly of the product.

Supplementary information on maintenance and preservation:

Change intervals for fluids and lubricants and details on preservation can be obtained in the relevant specifications of the component manufacturers and MTU Fluids and Lubricants Specifications. The latest version is also available at: <http://www.mtu-online.com>, for engine-generator sets at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

This maintenance schedule does not apply to engines with an EPA/CARB emission certificate.

1.2 Correlation matrix for application group 2A

Load profile		Load factor %	Load indicator %	12 V 1600 R50				
Load %	Time %			0 - 58 kW/cyl.				
110 ₍₁₀₀₋₁₁₀₎	—	0 - 18	≤1	30,000	—	—	—	Shunter Load Profile
100 ₍₉₀₋₁₀₀₎	0.5							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	4.5							
50 ₍₃₀₋₅₀₎	10							
30 ₍₅₋₃₀₎	20							
Idle ₍₀₋₅₎	65							
110 ₍₁₀₀₋₁₁₀₎	—	19 - 26	≤7	27,000	—	—	—	Multi-purpose Load Profile
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	15							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	20							
Idle ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	27 - 38	≤17	24,000	—	—	—	Standard Load Profile
100 ₍₉₀₋₁₀₀₎	10							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	15							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	39 - 43	≤18	21,000	—	—	—	High-speed Load Profile
100 ₍₉₀₋₁₀₀₎	15							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	25							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	20							
Idle ₍₀₋₅₎	40							
Max. number of load cycles 1				(<10% to >90% set value):5/h				
Max. number of load cycles 2				(<10% to >70..90% set value):10/h				
Max. number of starts				0.5 per engine operating hour				

Table 2: Correlation matrix for application group 2A

1.3 Maintenance schedule matrix 30,000 h

0 to 29,000 Operating hours

[illegible]

30,000 to 30,000 Operating hours

[illegible]

Task	Limit	Operating hours [h]																													
		30,000																													
W1675	2 a	X																													
W1019	-	X																													
W1207	2 a	X																													
W1013	2 a																														
W1326	2 a																														
W1005	3 a	X																													
W1525	6 a	X																													
W1526	6 a	X																													
W1776	-	X																													
W1042	-	X																													
W1777	6 a	X																													
W1250	6 a	X																													
W1251	6 a	X																													
W1298	9 a	X																													
W1006	9 a	X																													
W1636	9 a	X																													
W1178	9 a	X																													
W1041	9 a	X																													
W3148	18 a	X																													
w = weeks m = months a = years																															